

ToSCA: Leveraging Hierarchical Reinforcement Learning on Temporal and Strategic Abstractions of Conversational Agents

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Abstract

Humans have multiple levels of temporal abstractions on daily interaction and thinking, such as concept perception and strategic planning. Inspired by this nature, we propose a two-level hierarchical reinforcement learning (RL) framework for conversational agents, bridging the gap between previous token-level or utterance-level RL methods. Developed on a two-level MDP, the token-level response decoding is conditioned on the utterance-level action, the explicit textual strategies. Based on theoretical derivation and efficiency consideration, we use DQN to solve the high-level critic and PPO to solve the low-level actor-critic. To further alleviate the reward sparsity and facilitate the convergence, we also design the dual-granularity reward mechanism, in which the utterance-level satisfaction score is integrated with token-level intrinsic motivation and K-L penalty. Experiments on both daily and emotional support conversations show that our method outperforms versatile baselines in strategy determination and response quality. Our implementation is available at <https://github.com/AaronJi/ToSCA>.

1 Introduction

Large Language Models (LLMs) have made remarkable advances in conversational generation and reasoning (Ouyang et al., 2022; AI@Meta, 2024). However, when dealing with conversations, LLMs may encounter substantial obstacles: i) *the bottleneck of data supervision*, ii) *alignment on the nuance and complicated human preference*, as well as iii) *the wide distribution of topics*, challenging the adaptation and generalization of traditional finetuning methods (e.g., SFT or LoRA) (Hedayatnia et al., 2022; Yue et al., 2025; Lee et al., 2026). These issues motivate the adoption of reinforcement learning from human feedback (RLHF) for

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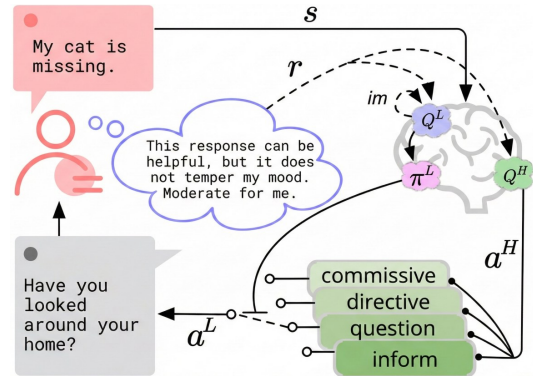
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Figure 1: The paradigm of ToSCA with two-level MDP. A high-level critic Q^H determines the optimal strategy (a^H), which guides the low-level actor π^L to produce the enhanced response (a^L). The strategic determination is depicted by a *switch* $\perp$ over the *contacts* $\multimap$. Q^H also bootstraps the user’s satisfaction reward, while the low-level critic Q^L also considers the intrinsic motivation. Case and strategies are sampled from DailyDialog.

conversation exploration and long-term return exploitation, such as Proximal Policy Optimization (PPO) (Ouyang et al., 2022) and Group Relative Policy Optimization (GRPO) (Shao et al., 2024).

Despite the success, in practice, most RL approaches operate at the token-level (Li et al., 2024a; Zhou et al., 2024b; Cho et al., 2024), suffering from reward sparsity and high computational overheads (Zhou et al., 2024b). These issues are more evident in multi-turn conversations, in which the meaningful user feedback is not available until the entire assistant utterance is completed (Li et al., 2024b). Consequently, exploration over the vast token-level action space becomes prohibitively expensive, preventing LLM-based agents from optimal, anthropomorphic conversational behaviors.

One possible remedy is inspired by humans’ multi-level abstractions capabilities (Murray et al., 2014; Zeraati et al., 2023). For example, humans often first decide a strategy, such as commissive, directive, question, and inform, then formulate

the detailed response. Such strategies can be extracted and annotated for human conversation corpus, based on sociological or psychological theories, such as the Helping Skills Theory (Hill, 2009) in ESConv (Liu et al., 2021), and the annotation criteria ISO 24617-2 (Petukhova et al., 2014) in DailyDialog (Li et al., 2017a). This hierarchical nature of human thinking motivates prompting methods (Zhao et al., 2026; Ning et al., 2024), supervised learning (Sun et al., 2023; Liu et al., 2024), retrieval-based framework (Yang et al., 2024), and simulation-based studies (Zhou et al., 2024a). Although recently there are also strategy-based RL methods such as StraQ* (Wang et al., 2025), they lack token-level action determinations, lacking the alignment of actions on strategy and token-levels.

Hierarchical Reinforcement Learning (HRL) (Sutton et al., 1999; McGovern and Barto, 2001) has addressed analogous challenges in robotics and control through temporal abstraction and action space reduction (Shi et al., 2025; Sun et al., 2025; Watanabe et al., 2025). However, due to fundamental differences in state and action representations, it is difficult to transfer traditional HRL methods to human conversations directly. Although recent studies also apply HRL on dialogue systems (Zhou et al., 2024b; Cho et al., 2024), to explore the dual-granularity of utterance-level and token-level generations, they may still have deficiencies. For example, the high-level policy of ArCher (Zhou et al., 2024b) determines the abstract intent in a continuous latent space, lacking explicit textual expressions, resulting in limited interpretability. In contrast, utterance-level RL approaches such as DAT (Li et al., 2024a) directly leverage explicit strategies to train a planner, while neglecting the rich demonstrations on the token level.

To bridge the gap between ArCher and DAT, in this paper, we propose a novel dual-level HRL framework called **Temporal-Strategic Abstractions of Conversational Agents (ToSCA)**, to guide a token-level policy with an utterance-level strategic planner. ToSCA belongs to a famous HRL paradigm called *options* (Sutton et al., 1999; Stolle and Precup, 2002), with the low-level action (response tokens) conditioned on the high-level action (strategies), as visualized in Figure 1. By defining a two-level MDP, we implement the high-level MDP with a value-based critic, operating on a discrete action space, solved by DQN; and the low-level MDP with the actor-critic framework, solved by PPO. To alleviate the reward sparsity and enhance the intrinsic

consistency, we also design a dual-granularity reward mechanism, in which the low-level reward has two token-level auxiliary components, the K-L penalty and the intrinsic motivation, besides the utterance-level user satisfaction score. We train and evaluate ToSCA on two domains, daily-life conversation (DailyDialog) and emotional support conversation (ESConv), as well as an out-of-domain test on EmpatheticDialogues. ToSCA surpasses prompting, supervised, and RL baselines, showing good generalization and reasonable strategic planning. Major contributions of this paper include:

- We propose a novel HRL-based dialogue system, bridging the gap between explicit utterance-level strategic planning and token-level response generation.
- We introduce intrinsic motivation into conversation generation, formulating a dual-granularity reward mechanism, to alleviate reward sparsity and facilitate the convergence.
- We design substantial experiments on both daily-life and emotional support conversations to verify the method’s effectiveness, and validate its generalization on out-of-domain tests.

2 Preliminaries

2.1 Markov Decision Process

The Markov decision process (MDP) is usually defined as a 5-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \gamma)$, where $\mathcal{S}$ is the state set, $\mathcal{A}$ is the action set, $\mathcal{R}$ is the reward set, γ is the discounting factor of rewards, and $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the state transition function.

At a specific time t , Reinforcement Learning (RL) can be employed to provide a policy $\pi(a|s)$ by optimizing the following discounted cumulative expected return (J):

$$\max_{\pi} \mathbb{E}_{s, a \sim \pi} \sum_{t=0}^{\infty} \gamma^t r_t := J(\mathcal{M}) \quad (1)$$

$$s \in \mathcal{S}, a \in \mathcal{A}, r \in \mathcal{R}$$

2.2 Q-Learning

Q-Learning belongs to value-based RL, which aims to learn the optimal state-action value function $Q^*(s, a)$, such that the determined action approximately optimizes J by maximizing Q :

$$a^* = \arg \max_a Q^*(s, a) \leftarrow \arg \max J$$

$$Q(s, a) \leftarrow r(s, a) + \gamma \max_{a'} Q(s', a') \quad (2)$$

in which the second equation is the famous Bellman Equation, the superscript $*$ means the optimal value, and $'$ indicates the next step.

Instead of explicitly implementing the above equation, Deep Q-learning (DQN) (Mnih et al., 2015) approximates the minimization of the error of Equation 2 with deep value networks:

$$\mathcal{L}_Q(\phi) = |r(s, a) + \gamma \max_{a'} Q_{\bar{\phi}}(s', a') - Q_{\phi}(s, a)|^2 \quad (3)$$

where ϕ and $\bar{\phi}$ are parameters of the Q and target Q-net, respectively. $\bar{\phi}$ is usually periodically synchronized from ϕ .

2.3 Proximal Policy Optimization

Proximal policy optimization (PPO) (Schulman et al., 2017) follows the Actor-Critic framework and is widely applied to LLMs (Ouyang et al., 2022). Initialized from an LLM backbone, the *actor* behaves as a generative policy network, while a separate value network serves as the *critic*. The actor is trained by maximizing the objective

$$\begin{aligned} \mathcal{L}_{\pi}(\theta) = & \mathbb{E}_{(s,a) \sim \mathcal{T}, \pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} A(s, a) \right] \\ & - \lambda_1 \mathbb{E}_{s \sim \mathcal{T}} \left[D_{\text{KL}}(\pi_{\theta}(\cdot|s) \parallel \pi^{\text{SFT}}(\cdot|s)) \right] \\ & + \lambda_2 \mathbb{E}_{(s,a) \sim \mathcal{D}_{\text{pretrain}}} \left[\log \pi_{\theta}(a|s) \right] \end{aligned} \quad (4)$$

where θ and θ_{old} denote the parameters of the current policy and the behavior policy, respectively, and $A(s, a)$ is the advantage function estimated by the critic. λ_1 controls the strength of the KL regularization that constrains the policy to remain close to the SFT model, while λ_2 balances the auxiliary pretraining objective. The critic is trained to regress the state-value function using

$$\mathcal{L}_Q(\psi) = \mathbb{E} \left[(Q_{\psi}(s) - A(s, a) - Q_{\bar{\psi}}(s))^2 \right] \quad (5)$$

where ψ and $\bar{\psi}$ are the parameters of the current and previous value networks, respectively.

3 Method

3.1 Task Formulation

Multi-turn conversation. A T -turn user-agent conversation can be represented by an interleaved sequence of $query(t)$ and $response(t)$, $1 \leq t \leq T$. This paper studies strategy-based conversations (Liu et al., 2021; Rashkin et al., 2019),

Level	Variable	Definition
High	$s^H \in \mathcal{S}^H$	State (<i>query, history, emo</i>)
	$a^H \in \mathcal{A}^H$	Strategy selection (<i>stra</i>)
	$r^H \in \mathcal{R}^H$	Satisfaction score (r^{sat})
Low	$s^L \in \mathcal{S}^L$	State ($s^H, response_i$)
	$a^L \in \mathcal{A}^L$	Current decoded token
	$r^L \in \mathcal{R}^L$	Token-level reward

Table 1: Variable definitions in the HRL framework.

with each turn annotated with the user’s emotion (*emo*) and the agent’s strategy (*stra*). Such an augmented conversation is then described by $\{desc, hist(T), query(T), emo(T), stra(T), response(T)\}$, in which *desc* is the session-level description, and $hist(T) := \{query(t), response(t)\}_{0:T-1}$ is the history context at t .

Hierarchical MDP. Similar to ArCher (Zhou et al., 2024b), we define the task of multi-turn conversations as a hierarchical MDP, conducting the temporal abstraction:

1. The high-level MDP $\mathcal{M}^H$: the agent makes conceptual decisions at the utterance level, guiding the low-level action.
2. The low-level MDP $\mathcal{M}^L$: Similar to conventional RLHF, this MDP operates at the token level, where the agent generates the response token by token, until completion.

where we use the superscripts H and L to represent the high and low levels. Different from Zhou et al. (2024b), our high-level action performs the strategic abstraction, which can be learned and verified by strategy-annotated datasets.

3.2 System Configuration

Variable definitions. According to the environment, we define its system variables as Table 1, where $response_i$ is the current generated part of *response*. Based the above definitions, $\mathcal{M}^H$ and $\mathcal{M}^L$ can be depicted with tuples of $(\mathcal{S}^H, \mathcal{A}^H, \mathcal{R}^H)$ and $(\mathcal{S}^L, \mathcal{A}^L, \mathcal{R}^L)$, respectively.

RL components and objectives. In this work, we employ HRL to jointly optimize the targets of $\mathcal{M}^H$ and $\mathcal{M}^L$. Different from previous configurations such as Actor^H-Critic^L (Zhou et al., 2024b), Actor^H-Critic^H-Actor^L (Bacon et al., 2017) or double Actor-Critic (Zhang and Whiteson, 2019), the high-level MDP employs a single critic (Q^H) to

provide a strategic bootstrapping on discrete options, while the low-level MDP inherits the actor-critic framework (π^L, Q^L) , taking advantage of robustness of token-level PPO. Therefore, our architecture can be named as *Critic^H-Actor^L-Critic^L*. All these components are initialized from the same pretrained LLM. ToSCA can be considered as the following objective:

$$\pi^H \leftarrow \max J(\mathcal{M}^H); \pi^L, Q^L \leftarrow \max J(\mathcal{M}^L) \quad (6)$$

where definitions inherit from Section 2.

3.3 The High-Level MDP

Training. Instead of augmenting the value head (as is done in conventional RLHF), we implement this strategic value function in a purely generative manner. Assuming there is an instruction template with the placeholder of s , denoted by $\mathcal{I}(s)$. Then the state-action value can be obtained from the averaged logits of the action tokens, from the conditional forward pass of LLM:

$$Q_\phi^H(s, a) \leftarrow \text{LLM}_\phi(\mathcal{I}^H(s) \oplus a) \quad (7)$$

where $\oplus$ represents the textual concatenation. We conduct DQN (Equation 3) to train Q_ϕ^H .

Prompt. We briefly exhibit our instruction $\mathcal{I}(s)$ here:

The high-level prompt $\mathcal{I}^H$

Description: {desc} **User's emotion:** {emo}
History: {hist} **Query:** {query}
Please select the best strategy:
(1) {stra 1} **(2)** {stra 2} $\dots$ **(K)** {stra K}

with a full version in Appendix A. To further strengthen the understanding capability of LLM on the strategy selection, we formulate $\mathcal{I}$ as a multiple-choice question (MCQ), instead of a plain question, forcing the LLM to choose one of the options. Accordingly, the action set becomes the set of possible strategy indices $a \in \mathcal{A} := \{1, 2, \dots, K\}$ where K is the total number of strategies.

Inference. Instead of decoding the next token, the finetuned LLM produces logits of available strategies, and the optimal strategy can be determined from the maximum logit

$$a^H \leftarrow \arg \max \text{LLM}_\phi(\mathcal{I}^H(s) \oplus a), a \in \mathcal{A} \quad (8)$$

3.4 The Low-Level MDP

Training. Another LLM backbone is employed as the low-level policy π^L , with the trainable parameter θ . Similar to RLHF (Ouyang et al., 2022), a copy of LLM backbone is used as the low-level critic Q^L , with the value head augmented. We use token-level PPO to train π^L and Q^L jointly.

Inference and prompt. The high-level action guides the low-level action by in-context learning (ICL):

$$a^L \leftarrow \text{LLM}_\theta(\mathcal{I}^L(s^L, a^H)) := \pi_{a^H}^L(s^L) \quad (9)$$

The low-level prompt $\mathcal{I}^L$

Description: {desc} **User's emotion:** {emo}
History: {history} **Query:** {query}
Your strategy: {stra} **Please response now:**

The prompt has the above template with detailed content in Appendix A.

3.5 The Reward Mechanisms

To provide multi-timescale feedback, we define our high-level reward (r^H) as the satisfaction score (r^{sat}) at the real time t , solely coming from the environment; while the low-level reward (r^L) has another two token-level (k) components

$$r_t^H = r_t^{\text{sat}}, \quad r_k^L = r_k^{\text{sat}} + \beta_1 r_k^{\text{KL}} + \beta_2 r_k^{\text{im}} \quad (10)$$

with definitions of reward components below.

The satisfaction score (r^{sat}). Choice of rewards may be crucial, especially when an offline dataset constrains the sampling. In this paper, we leverage the LLM-as-the-Judge approach to automatically generate online scores. A strong-basis LLM, such as GPT-4o, is employed as the oracle (or the teacher) to produce the satisfaction score r^{sat} from 0 to 5. In Appendix A, we provided the detailed evaluation prompt; in Appendix F, we conduct statistical experiments to illustrate the consistency between oracle and human scores.

The KL reward (r^{KL}). The penalty of Kullback–Leibler (K-L) divergence prevents π^L from deviating the ref policy too much, which has the form of $r_{\text{KL}} = -D_{\text{KL}}(\pi^L \parallel \pi^{\text{ref}})$.

Intrinsic motivation (r^{im}). Here we define the reward of intrinsic motivation as the inner-agent alignment between the low-level context (state plus strategy) and response:

$$r^{\text{im}} = \log \text{LLM}_\theta(s^L \oplus a^H \oplus a^L) \quad (11)$$

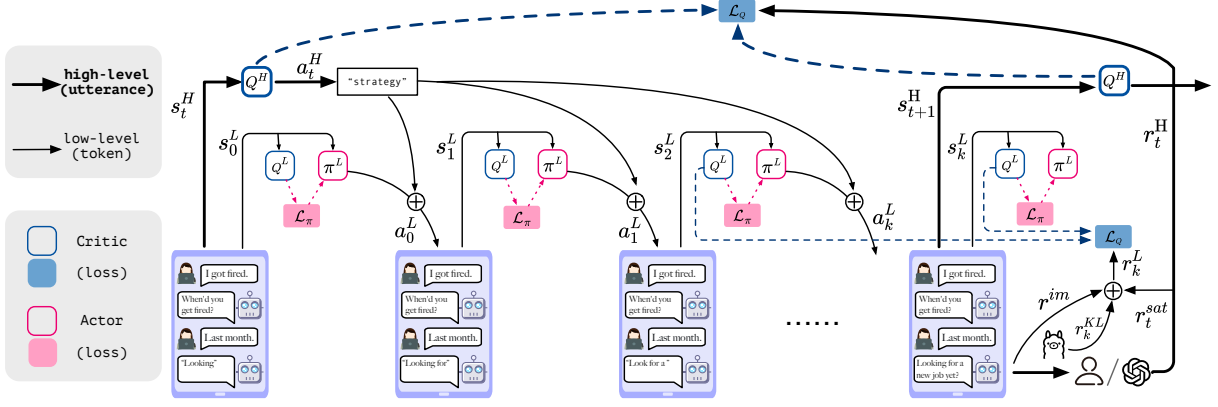


Figure 2: Illustration of the dual-scale pipeline of ToSCA. The high-level MDP operates on the utterance scale (t), receives the user’s satisfaction reward ($r_t^H = r_t^{sat}$), and determines the *strategy* (a_t^H) which conditions the low-level action (a_k^L). The low-level MDP operates on the token scale (k), learned by the actor-critic framework. The low-level reward (r_k^L) consists of the satisfaction reward (r_k^{sat}), KL penalty (r_k^{KL}) and intrinsic motivation (r_k^{im}). Q^H and Q^L are optimized by the critic loss $\mathcal{L}_Q$, while π^H is optimized by $\mathcal{L}_\pi$.

where the right-hand side can also be observed as the cross-entropy loss of LLM with all textual inputs, similar to the pretrained loss in (Ouyang et al., 2022). By augmenting r^{im} , we encourage the low-level policy to be more proficient with the strategy-steering response style, in contrast with the policy update driven solely by human feedback.

3.6 Algorithm

Algorithm 1 outlines our overall framework. For each session, we sample an initial query from the dataset and perform the rollout. At each turn, the high-level critic first determines a strategy, then the low-level policy generates the response, conditioned on both state and strategy. During training, the high-level critic is learned by DQN, while the low-level actor and critic are trained by PPO. Figure 2 illustrates the entire pipeline. Further implementation details are provided in Appendix A.

3.7 Theoretical Derivation

We start from two assumptions proposed by the *options* framework (Sutton et al., 1999), which are the foundation of the subsequent main theorem.

Assumption 1. *The strategies, as special types of options, are everywhere:*

$$\forall s \in \mathcal{S}, \forall a^H \in \mathcal{A}^H : \exists s \in \mathcal{I}(s|a^H) \quad (12)$$

where $\mathcal{I}$ is the initial state set.

Assumption 2. *The strategies are Markovian:*

$$P(s_t^H, a_t^H | s_{0:t-1}^H, a_{0:t-1}^H) = P(s_t^H, a_t^H | s_{t-1}^H, a_{t-1}^H) \quad (13)$$

Considering the features of the dialogue system, it is obvious to see both assumptions are reasonable. The agent can choose any strategy or any response, no matter what the current state is. The prior history and query strategy are also prerequisites for determining the next strategy. Within a hierarchical framework, we propose the main theorem:

Theorem 1. *With Assumptions 1 and 2 hold, the value function (Eq 7) converges as long as the high-level policy (Eq 8) is deterministic, and the high- and low-level critics are solved interleaved.*

Due to page limits, we leave detailed derivations in Appendix E. Theorem 1 suggests that our determined strategy can converge to the ground truth.

4 Experiment

In this section, we mainly address two research questions (RQ): **RQ1**, whether our framework can adapt to different conversation domains; **RQ2**, whether our method can generalize well when strategy annotation is not available. For **RQ1**, we investigate two domains in Section 4.5 and 4.6, respectively; for **RQ2**, we differentiate the tests into in-domain and out-of-domain tests, depending on the availability of strategy annotations.

4.1 Implementation

We use LLaMA3.2-1B-Instruct and LLaMA3.1-8B-Instruct (AI@Meta, 2024) as high-level and low-level backbones, respectively. The framework is trained on OpenRLHF (Hu et al., 2024) using 16 A100 GPUs. We employ GPT-4o for both user simulation and satisfaction scoring. We use learning

Algorithm 1 HRL-ToSCA

Input: Q^H, Q^L, π^L from pretrained checkpoints; batch sizes B^H and B^L

```

1: Initialize buffer  $\mathcal{B}^H, \mathcal{B}^L = \{\}$  .
2: while not converges do
3:   Initialize  $s^H, s^L$  with queries from dataset
4:   \Roll-out
5:   for  $t = 1, \dots, T$  do
6:     Inference  $a_t^H$  by arg max (Eq 8)
7:     for  $k = 1, \dots, K$  do
8:       Inference  $a_t^L \sim \pi^L(s_t^L)$  (Eq 9)
9:       Increment  $s_t^L$  to  $s_{t+1}^L$ 
10:      break if  $a_t^L = < EOS >$ 
11:    end for
12:    collect  $r_t^{\text{sat}}$  from the oracle LLM
13:    Append buffer  $\mathcal{B}^L \cup \{(s_k^L, a_k^L, r_t^{\text{sat}})\}_{k=1}^K$ 
14:    Append buffer  $\mathcal{B}^H \cup (s_t^H, a_t^H, r_t^{\text{sat}})$ 
15:  end for
16:  \High-level Training
17:  Sample  $(s_t^H, a_t^H, r_t^{\text{sat}}, s_{t+1}^H, a_{t+1}^H)$  from  $\mathcal{B}^H$ 
18:  update  $Q^H$  by DQN (Eq 3)
19:  \Low-level Training
20:  for  $k = 1, \dots, K - 1$  do
21:    Get  $(s_k^L, a_k^L, r_t^{\text{sat}}, s_{k+1}^L, a_{k+1}^L)$  from  $\mathcal{B}^L$ 
22:    Calculate  $r_k^{\text{KL}}, r_k^{\text{im}}, r_k^L$  by Eq 10
23:    Update  $\pi^L$  and  $Q^L$  jointly by PPO (Eq 4)
24:  end for
25: end while

```

rates of 5e-6, 9e-7, and 9e-4 for the training of Q^H , π^L , and Q^L , respectively. The batch size is 64, and the window length is set to 128.

4.2 Dataset

Daily-life conversations. DailyDialog (Li et al., 2017a) is a manually constructed multi-turn dialogue dataset with approximately 13,000 conversations on everyday topics. Each utterance is annotated with a dialogue strategy¹, including *Question*, *Inform*, *Directive*, and *Commissive*. During training, only the strategy set is provided, and the model infers strategies without access to gold labels. The annotations are reserved exclusively for the evaluation of strategy classification and response quality.

Emotional support conversations (ESC). Classical datasets in this field include ESConv (Liu et al., 2021), annotating with 8 strategies; and EmpatheticDialogues (Rashkin et al., 2019), which has **no** strategy annotation. As a result, we employ

¹Denoted as ‘Act’ in the original dataset.

Method	$< Q^H >$	$< r^L >$
Direct	486.2	3.21
Raw Dataset	551.4	3.53
ToSCA (ours)	616.5	4.03

Table 2: Averaged values and rewards on DailyDialog.

ESConv as the in-domain (ID) test while EmpatheticDialogues as the out-of-domain (OOD) test. Appendix B provides more detailed introductions and statistics of these datasets.

4.3 Evaluation

Classification metrics. We employ classification metrics of accuracy (Acc) and Macro-F1 (MaF1). We also refer to the evaluation methods proposed by Kang et al. (2024), which propose the *bias* of strategies based on the Bradley-Terry model (Bradley and Terry, 1952). A smaller *bias* indicates better.

Automatic metrics on response. For generative tasks, we utilize similarity-based metrics like Bleu-2 (B-2) and Rouge-L (R-L); as well as Distinct-2 (D-2), which indicates response diversity.

Human scoring. Similar with Kang et al. (2024), we annotate with dimensions of *Acceptance*, *Effectiveness*, *Sensitivity*, *Fluency*, and *Emotion*, and the ultimate purpose, seeker’s *Satisfaction*. Detailed scoring settings and principles are in Appendix C.

4.4 Baselines

We consider several types of baselines, including (1) Direct inference; (2) Prompting methods such as Self-Refine (Madaan, 2023), ECoT (Li et al., 2024c), SoT (Ning et al., 2024), ToT (Yao et al., 2023), Plan-and-Solve (PS) (Wang et al., 2023), Finite State Machine (FSM) (Wang et al., 2024); (3) finetuning methods such as SFT, and the supervised versions of ECoT and FSM (EmoFSM) (Zhao et al., 2026); and 4) RL methods such as DQN and PPO, and two HRL baselines: DAT (Li et al., 2024b) and ArCher (Zhou et al., 2024b). Detailed introductions and implementation details of baselines can be found in Appendix D.

4.5 Results on Daily-Life Conversations

Reward and Value. Loss convergence and reward&value increase are observed in our training. Table 2 shows the averaged rewards and values of ToSCA on DailyDialog, compared to Direct and Raw dataset. Loss curves are in Appendix F.

Methods	Acc $\uparrow$	MaF1 $\uparrow$	<i>bias</i> $\downarrow$	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
Direct	52.60	18.03	1.66	3.35	10.33	44.74
+ Direct-Refine	48.27	28.28	0.70	2.56	8.70	43.67
+ Self-Refine (Madaan, 2023)	49.76	22.15	1.18	2.40	7.75	34.01
+ ECoT (Li et al., 2024c)	38.94	29.99	0.27	1.78	6.00	55.26
+ SoT (Ning et al., 2024)	N/A	N/A	N/A	2.53	7.97	59.98
+ ToT (Yao et al., 2023)	N/A	N/A	N/A	2.52	8.84	43.19
+ PS (Wang et al., 2023)	N/A	N/A	N/A	2.60	7.76	39.73
+ FSM (Wang et al., 2024)	46.86	21.22	1.30	2.70	9.44	38.75
SFT	60.19	44.82	0.82	6.81	18.52	43.36
+ECoT	60.11	44.9	0.66	6.61	18.07	42.87
+FSM (Zhao et al., 2026)	60.03	46.02	0.55	5.85	21.77	47.43
DQN	54.01	50.10	0.62	4.18	13.09	59.27
DAT (Li et al., 2024b)	N/A	N/A	N/A	3.45	11.80	0.90
ArCher (Zhou et al., 2024b)	50.41	42.67	0.21	5.17	14.35	55.16
ToSCA (ours)	63.64	58.91	0.63	16.35	35.22	62.67

Table 3: ID results of automatic metrics including Acc, MaF1, *bias*, B-2, R-L and D-2 on DailyDialog. The best result of the methods is **bolded**.

Strategy determination. In Table 3, although ToSCA does not receive reward signals from the ground-truth strategy labels, it still achieves the Acc, benefiting from the alignment between human-annotated data and the automatic evaluation of correct strategies. While ToSCA does not have the best *bias* score, it performs competitively, which collectively leads to the best MaF1 of 58.91.

Response quality. Table 3 also illustrates that ToSCA produces responses with the highest similarity to ground-truth (as measured by B-2 and R-L), while simultaneously achieving the best diversity (D-2), surpassing prompt, finetuning, or RL-based baselines, as well as specific structured models on DailyDialog. More detailed results (including full lists of baselines) are in Appendix F.

4.6 Results on ESC

In-domain results. In Table 4, as the scenario shifts from daily dialogues to emotional support conversations, ToSCA achieves the best or second-best performance across all metrics, demonstrating its ability to employ appropriate empathetic strategies and achieve better emotional alignment. We further conduct human evaluation in Appendix F.3; as well as comparisons to previous dataset-specific studies (Li, 2020; Farahani, 2023; Kang et al., 2024) in Appendix F.4. Typical cases are in Appendix F.7.

Transfer to OOD. In this setting, we omit the finetuning baselines to mimic the knowledge transfer scenario. Table 5 demonstrates the generalization ability of ToSCA on EmpatheticDialogues, with higher performance than the baselines on this

Methods	Acc $\uparrow$	MaF1 $\uparrow$	<i>bias</i> $\downarrow$	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
Direct	11.80	10.26	1.61	3.47	10.64	33.45
+ Direct-Refine	17.08	11.07	1.27	3.10	6.13	14.22
+ Self-Refine (Madaan, 2023)	17.58	13.61	1.92	3.34	9.71	14.61
+ ECoT (Li et al., 2024c)	15.32	10.38	1.69	3.16	10.50	30.38
+ SoT (Ning et al., 2024)	N/A	N/A	N/A	3.07	8.76	26.15
+ ToT (Yao et al., 2023)	N/A	N/A	N/A	2.65	9.81	14.2
+ PS (Wang et al., 2023)	N/A	N/A	N/A	2.81	8.27	19.12
+ FSM (Wang et al., 2024)	17.37	11.15	0.81	4.12	11.83	35.43
SFT	32.43	21.29	1.28	6.97	16.59	50.45
+ECoT	30.80	17.70	1.35	6.51	15.00	34.96
+FSM (Zhao et al., 2026)	28.00	23.70	0.41	5.88	15.30	51.48
DQN	<u>37.69</u>	<u>34.57</u>	0.59	3.59	11.74	44.14
DAT (Li et al., 2024b)	18.50	8.36	1.74	3.26	11.24	35.82
ArCher (Zhou et al., 2024b)	24.50	19.60	0.50	5.30	13.10	54.80
ToSCA (ours)	39.26	36.85	<u>0.48</u>	<u>6.93</u>	<u>16.28</u>	<u>52.42</u>

Table 4: ID results of automatic metrics including Acc, MaF1, *bias*, B-2, R-L and D-2 on ESConv. The best result is **bolded** and the second best is underlined.

zero-shot test. This success benefits from the inclusion of empathy elements in the reward signal derived from daily dialogues.

Methods	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
Direct	3.09	9.91	25.23
+ ECoT(Li et al., 2024c)	2.91	9.79	32.65
+ SoT (Ning et al., 2024)	1.79	5.66	48.59
+ ToT (Yao et al., 2023)	2.31	9.05	29.09
+ PS (Wang et al., 2023)	2.69	6.93	24.02
+ FSM (Wang et al., 2024)	3.33	10.80	33.37
DQN	3.61	9.15	39.60
DAT (Li et al., 2024b)	3.38	<u>10.90</u>	35.05
ArCher (Zhou et al., 2024b)	<u>3.84</u>	10.75	45.19
ToSCA (ours)	4.49	12.93	<u>46.53</u>

Table 5: OOD results of B-2, R-L and D-2 on EmpatheticDialogues. The best result of the methods is **bolded** and the second best is underlined.

Ablation Study. Here we conduct three ablation studies: (1) *w/o high-level*, which degenerates to token-level PPO by removing the high-level critic; (2) *w/o low-level*, which employs a DQN-based strategy planner (*i.e.*, StraQ* (Wang et al., 2025)) to guide the token generation of a frozen LLM; and (3) *w/o im*, which removes the intrinsic motivation from the low-level reward. Table 6 shows that removing either the high-level or low-level component results in a significant performance drop, verifying the hierarchical design. *w/o im* also causes a moderate but evident decrease in automatic metrics, indicating the effectiveness of intrinsic motivation. Appendix B further presents sensitivity analyses on γ and reward weights (β_1 and β_2).

Method	Acc $\uparrow$	MaF1 $\uparrow$	bias $\downarrow$	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
w/o high-level	N/A	N/A	N/A	7.85	25.16	50.59
w/o low-level	53.39	49.02	0.53	4.17	13.15	59.44
w/o im	60.04	53.01	0.68	16.21	35.06	60.95
ToSCA (ours)	63.64	58.91	0.63	16.35	35.22	62.67

Table 6: Ablation study of ToSCA on DailyDialog.

4.7 Discussion

Method variance across domains. From Table 3 and Table 4, one can observe difference performances for several baselines. For example, SFT and ArCher perform similarly or even outperform ToSCA on ESConv, yet perform poorly on DailyDialog. This difference may be because ESConv falls within the ESC domain, with limited topics and clear objectives, resulting in relatively well-defined behavior patterns (Exploration->Comforting->Action, as in (Liu et al., 2021)) which are easier for supervised methods to imitate. On the contrary, DailyDialog has open-ended topics, more spontaneous emotional exchanges, and much broader behavior distributions, which may challenge pure imitations. Furthermore, the high-level policy of ArCher maps abstract intents into a continuous latent space, lacking explicit textual expressions (e.g., “Questioning” or “Acknowledging” in ToSCA), leading to a deficiency in semantic alignment. Despite these obstacles, our ToSCA has robust performance across all three datasets, indicating its strong generalization and adaptation.

Intrinsic motivation of strategic chatting. In ToSCA, high-level strategies guide the low-level policy via prompts, but fine-tuning may cause the model to ignore these constraints for higher rewards. As shown in Table 7, without intrinsic motivation, the model fails to follow the *Question* strategy. Intrinsic motivation enforces strategy-aligned behavior during multi-turn training. Additional examples are in Appendix F.

User	It’s a portable TV. It’s a popular thing now.
Assistant	Oh, that’s new to me. So what’s on everyday?
User	News about current affairs, documentaries, music, movies, noncommercial ads and so on.
w/o im	(Question) I bet there’s something fun to watch on it.
w/ im	(Question) Got anything exciting on there?

Table 7: Comparison of responses with and without intrinsic motivation (im). The response without intrinsic motivation (w/o im) shows lower self-consistency.

Lookahead strategy determination. To better understand the strategy determination, we also investigate the strategy transitions, with each grid cell (i, j) representing the frequency from the strategy on i -th row to the strategy on j -th column.

Figure 3 (Left) exhibits the normalized occurrences of strategy distributions from user to ToSCA in the same turn. Clearly, “*Question* $\rightarrow$ *Inform*” and “*Directive* $\rightarrow$ *Commissive*” are the two most frequent transitions, which are consistent with the conversation logic, indicating the agent aims to answer the user’s question or provide help.

Figure 3 (Right) then shows how ToSCA’s strategies transition between adjacent utterances of itself, revealing ToSCA’s lookahead capability on strategy abstraction. The upper-right triangle suggests a dialogue pattern statistically: *Inform* $\rightarrow$ *Question* $\rightarrow$ *Directive* $\rightarrow$ *Commissive*, which shows the agent’s strategy pipeline, consistent with the Helping Skills Theory discussed in Liu et al. (2021).

5 Related Work

5.1 Hierarchical Reinforcement Learning

Hierarchical Reinforcement Learning (HRL) structures hierarchical MDPs via multi-level decision-making, mainly including *goal-conditioned* frameworks (Kulkarni et al., 2016; Vezhnevets et al., 2017; Levy et al., 2019) and *options*-based frameworks with discrete subskills (Bacon et al., 2017; Chunduru and Precup, 2020; Zhang and Whiteson, 2019). Our framework belongs to the latter with the semantic connections between two-level actions.

5.2 Utterance-level RL for Dialogue Systems

Utterance-level RL extends token-level RLHF (Ouyang et al., 2022) to alleviate reward sparsity. DAT models utterance actions in a continuous space (Li et al., 2024a), while ArCher adopts a two-level HRL framework with a learned high-level critic (Zhou et al., 2024b). Our method follows a two-level HRL structure but introduces discrete strategy abstraction at the high level.

6 Conclusion

In this paper, we propose ToSCA, an HRL-based dialogue agent with temporal and strategic abstractions. It includes a high-level critic that produces strategy-level actions, guiding the low-level policy to generate the detailed response. To ensure sampling efficiency and convergence, we train the high-level critic by DQN, and train the low-level

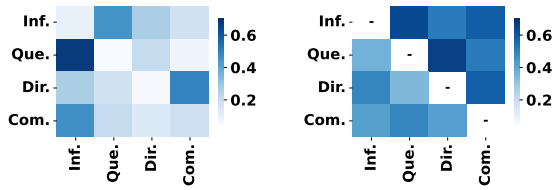


Figure 3: Occurrences of strategy (in abbrev.) transitions. *Left*: Strategy transitions from user (row) to ToSCA (column). *Right*: Strategies transitions of ToSCA between adjacent turns (row $\rightarrow$ column).

actor and critic by PPO. To improve the semantic alignment between the dual-level actions, we further incorporate a KL penalty and intrinsic motivation reward. Experiments on daily-life and emotional support conversations, as well as the OOD test, demonstrate the effectiveness of ToSCA.

Limitations

Compared to conventional supervised and RL methods, the hierarchical framework introduces additional training complexity, which may require careful tuning to ensure stable convergence. Furthermore, this study mainly focuses on conversations, while it would also be promising to develop a generalized, self-evolving hierarchical reasoning and thinking AI.

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A Implementation Details

A.1 Detailed Prompts

Prompt of the high-level agent. The following is a high-level prompt. In DailyDialog, there are 4 predefined strategies. With this prompt, the language model is naturally inclined to generate one of the candidate options as the next token due to the structure of the instruction.

High-level prompt

You are given a multi-turn dialogue between a user and an assistant. The user’s basic situation is as follows:

Emotion: $\{emo\}$

Description: $\{desc\}$

Below is the dialogue history between the user and the assistant:

$\{history\}$

The user’s current query is:

$\{query\}$

Based on the above context, please select the most appropriate response strategy from the following options:

strategy #(1) $\{a_1\}$

...

strategy #(k) $\{a_k\}$

Please provide your selection in the format of (1) through (k). Your selection is:

Prompt of the low-level agent. The low-level prompt derives the response strategy from the high-level and generates an appropriate reply accordingly.

The evaluation prompt of GPT-4o. The responses generated by the low-level prompt are evaluated by GPT-4o to simulate human judgment across five dimensions: *Acceptance*, *Effectiveness*, *Sensitivity*, *Fluency*, and *Emotion*. Each dimension is scored as either 0 or 1, indicating whether the response is strong in that aspect. The final output includes the total score and a brief explanation.

A.2 More Methodology Details

Kullback-Leibler (KL) Divergence is a fundamental concept in information theory and machine learning that measures the difference between two probability distributions. It is a crucial tool for comparing and approximating distributions, with far-reaching implications in various fields. The KL

Low-level prompt

You are given a multi-turn dialogue between a user and an assistant. The user’s basic situation is as follows:

Emotion: $\{emo\}$

Description: $\{desc\}$

Below is the dialogue history between the user and the assistant:

$\{history\}$

The user’s current query is:

$\{query\}$

The current response strategy is:

$\{stra\}$

Based on the current response strategy and other information, please act as a assistant and provide the best response. Keep replies brief without additional pronouns or extra elements.

Divergence between two probability distributions P and Q is defined as: s

$$D_{KL}(P||Q) = \sum_{i=1}^N p(x) \log \frac{p(x)}{q(x)} \quad (14)$$

where $p(x)$ and $q(x)$ are the probability density functions of P and Q respectively.

B Experiment Settings

B.1 Datasets

Detailed introductions. We introduce our experimental datasets in more details:

- The *DailyDialog* dataset (Li et al., 2017b) is a widely used benchmark for daily-life conversation. It contains 13,118 multi-turn dialogues across diverse daily topics (e.g., family, work, hobbies) with manually annotated emotions and dialogue strategies (named ‘act’ in the original dataset). As shown in Table 8, it is annotated with 7 emotions and 4 strategies: *Inform(Inf.)*, *Question(Que.)*, *Directives(Dir.)*, and *Commissive(Com.)*.
- The *ESConv* dataset (Liu et al., 2021) is a specialized corpus for emotional support research. It contains 1,000+ multi-turn dialogues where users articulate personal struggles (e.g., workplace stress, interpersonal conflicts, self-esteem issues), and support providers respond with empathy, validation,

Evaluation Prompt

You are an expert evaluator simulating human assessment of dialogue responses. You are given a multi-turn dialogue between a user and an assistant. The user’s basic situation is as follows:

Emotion: {*emo*}

Description: {*desc*}

Below is the dialogue history between the user and the assistant:

{*history*}

The user’s current query is:

{*query*}

The assistant’s response is:

{*response*}

Please evaluate the assistant’s response across five binary-scored dimensions.

1. *Acceptance* – Is the response socially appropriate and non-offensive?
2. *Effectiveness* – Does the response address the user’s intent appropriately?
3. *Sensitivity* – Does the response consider emotional or situational context?
4. *Fluency* – Is the response grammatically correct and fluent?
5. *Emotion* – Does the response convey an appropriate emotional tone?

Each dimension should be scored as **0 (unsatisfactory)** or **1 (satisfactory)**.

Output format:

Acceptance: [0/1], *Effectiveness*: [0/1]

Sensitivity: [0/1], *Fluency*: [0/1]

Emotion: [0/1], *Total score*: [0–5]

Explanation: [your reasoning here]

and actionable guidance. Aligned with Table 8, it is annotated with 11 emotions and 8 conversational strategies. The strategies are structured across three stages (I–III).

- The *EmpatheticDialogues* dataset (Rashkin et al., 2019) contains conversations centered on recognizing and responding to emotions (e.g., joy, sadness, frustration). It has annotations of user emotions but no strategies.

Statistics of datasets. Table 9 summarizes the statistics of ESConv, DailyDialog, and EmpatheticDialogues. It presents key metrics like the number of sessions, utterances, average utterance length, and counts of strategies and emotions for both

Dataset	Strategies	Abbr.	Stage
DailyDialog	Inform	Inf.	-
	Question	Que.	-
	Directives	Dir.	-
	Commissive	Com.	-
ESConv	Question	Que.	I
	Restatement or Paraphrasing	Res.& Par.	I
	Reflection of Feelings	Ref.	II
	Self-disclosure	Self-Dis.	II
	Affirmation and Reassurance	Aff.& Rea.	III
	Providing Suggestions	Pro.	III
	Information	Inf.	III
	Others	Others	-

Table 8: Strategies, abbreviations, and stages within DailyDialog and ESConv.

Category	ESconv	DailyDialog	Empathetic-
# Sessions	1.3K	13.1k	2.5K
# Utterances	38K	103.0k	11.0K
Average # Utterances	28.9	7.9	4.3
Average Utterance Length	18.8	13.6	16.7
Speaker1	# Utterances	20K	53.8k
	Avg # Utterances	15.4	4.1
	Avg Uttr Len	16.8	13.2
	# Strategies	-	4
	# Emotions	11	7
Speaker2	# Utterances	18K	49.2k
	Avg # Utterances	13.6	3.9
	Avg Uttr Len	21.0	14.1
	# Strategies	8	4
	# Emotions	-	7

Table 9: Statistics of ESConv, DailyDialog and EmpatheticDialogues (named Empathetic- in the table).

speakers in each dataset. For instance, DailyDialog has the most sessions (13.1k) and utterances (103.0k in total for both speakers), while ESConv shows detailed strategy annotations (11 for Speaker1) and a moderate number of sessions (1.3k). EmpatheticDialogues, while not annotated with strategies, offers a rich set of 32 emotion categories, making it valuable for studying emotional diversity and text-based evaluation.

In our experiment, we use the following data splitting strategy:

- *DailyDialog*: Following its official default configuration, the dataset is divided into a training set (11118 conversations), a validation set (1000 conversations), and a test set (1000 conversations).
- *ESConv*: We split the dataset into the training and test sets with a random 9:1 ratio, since the original dataset does not provide the official splits.
- *EmpatheticDialogues*: Following its official default configuration, Conversations are parti-

<i>Topic</i>	I hate my job but I am scared to quit and seek a new career.
<i>Query</i>	[<i>history</i>] <i>seeker</i> : Seriously! What I'm scared of now is how to secure another job.
<i>Emotion</i>	Anxiety (intensity: 5)
<i>Strategy</i>	Reflection of feelings
<i>Response</i>	<i>supporter</i> : I can feel your pain just by chatting with you.

Table 10: An example of ESConv.

tioned into approximate splits of 80% training (19533 conversations), 10% validation (2770 conversations), and 10% testing (2547 conversations). To avoid data leakage, all conversations sharing the same initial situational description (provided by a speaker) are grouped into the same partition.

Example of data samples. Table 10 shows an example dialogue snippet from the ESConv dataset. It illustrates a conversation where the seeker expresses anxiety about quitting a disliked job without a secure alternative. The dialogue is annotated with the topic, the seeker’s query, the emotional state (anxiety with high intensity), and the empathetic strategy used by the supporter—in this case, a “reflection of feelings.” This example highlights how ESConv captures nuanced emotional expression alongside supportive conversational strategies.

B.2 Key Hyperparameters

Full hyperparameter values. Table 11 shows the full list of hyperparameters in the training process of DailyDialog and ESConv. The training program is running on OpenRLHF, with zero stage

Group	Parameter	DailyDialog	ESConv
SFT	Lr	5e-5	5e-5
	batch size	64	64
	high-level:		
	Lr of Q^H	5e-6	5e-6
	batch size (B^H)	64	64
	replay buffer size	20000	20000
	max window len	1024	2048
	update freq of $\phi \rightarrow \bar{\phi}$	10	10
	γ	0.85	0.85
	low-level:		
ToSCA	Lr of π^L	9e-7	9e-7
	Lr of Q^L	9e-4	9e-4
	batch size (B^L)	64	64
	replay buffer size	20000	20000
	max window len	1024	2048
	max decoding length	128	128
	β_1	0.01	0.01
	β_2	0.01	0.01
	γ	1.0	1.0

Table 11: Full list of hyperparameters.

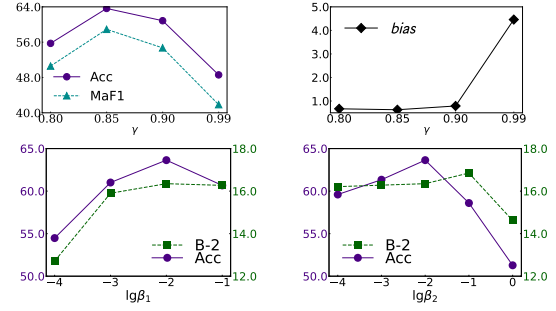


Figure 4: Sensitivity plots of ToSCA-imit on different γ and model sizes. Metrics include Acc, Q , B2, and R-L.

(of Deepseed) of 3, and the data type of bf16.

For the policy inference on all three datasets, we conduct the general decoding process of LLM, with top-p of 0.9, top-k of 10, and temperature of 0.7.

Sensitivity analysis on γ . Figure 4 (upper) shows ID (on *ESConv*) performance curves under different γ settings. One can observe the optimal γ happens on 0.85, which corresponds to higher **Acc** and **MaF1**, as well as lower **bias**. The curves of generative metrics have similar trends. Therefore, we choose $\gamma = 0.85$ as the formal setting.

Sensitivity analysis on reward weights. To further explore the effectiveness of the reward components added to the low-level reward (r^L), we conduct the sensitivity analysis on their weights, β_1 and β_2 . The results are shown on Figure 4 (low). First, the curves reveal that including of both loss terms (r^{KL} and r^{im}) have positive effects, with the performance increasing from less to larger values of weights. Second, we empirically find the optimal value of β_1 happens at 0.01, and $\beta_2 = 0.01$ has better strategy accuracy, while $\beta_2 = 0.1$ has higher $B - 2$ result. To make the setting simple, we temporally choose both $\beta_1 = \beta_2 = 0.01$ in the formal experimental setting.

C Detailed Evaluation Methods

C.1 Classification Metrics

Accuracy. We define evaluate strategy determination from the classification perspective, where the accuracy (Acc) is defined as the fraction of correctly predicted strategies over the entire number of strategies, comparing to the annotated strategies in the dataset:

$$\text{Acc} = \frac{\# \text{ of correctly predicted strategies}}{\# \text{ of annotated strategies}} \quad (15)$$

F1-scores. F1-related scores include Micro-F1 and Macro-F1. Micro-F1 considers the overall precision and recall of all instances, while Macro-F1 equals the average F1-score of labels.

bias. We define the preference *bias* as *how much the model prefers certain labels over others*. To quantify the preference for each strategy in LLMs, we employ the Bradley-Terry model (Bradley and Terry, 1952), which is widely used in human preference modeling (Rafailov et al., 2023). Following Newman (2023), we formally derive the preference p for strategy i as follows:

$$p'_i = \frac{\sum_j (w_{ij} p_j) / (p_i + p_j)}{\sum_j w_{ji} / (p_i + p_j)} \quad (16)$$

where w_{ij} represents the number of times the model predicts strategy i when the ground-truth strategy is j . All of the preference p_i are initialized as 1 and updated through iteration of the Eq (16), where p'_i represents the preference in the next iteration. After the final iteration, we scale the total sum of p_i to 8 ($\sum p_i = 8$) so that the average $\bar{p}$ becomes 1, indicating a strong preference for strategy i if $p_i > 1$.

We use a standard deviation of preferences p_i across the strategies as *bias*.

$$bias = \sqrt{\frac{\sum_{i=1}^N (p_i - \bar{p})^2}{N}} \quad (17)$$

where a higher value for *bias* indicates that the model exhibits a clear preference for both preferred and non-preferred strategies (Kang et al., 2024).

C.2 Generative Metrics

Bleu-2. B-2 (Papineni et al., 2002) first compute the geometric average of the modified n -gram precisions, p_n , using n -grams up to length N and positive weights w_n summing to one.

Next, let c be the length of the prediction and r be the reference length. The BP and Bleu-2 are computed as follows.

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases} \quad (18)$$

$$Bleu = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (19)$$

Rouge-L. R-L (Lin, 2004) propose using LCS-based F-measure to estimate the similarity between two summaries X of length m and Y of length n , assuming X is a reference summary sentence and Y is a candidate summary sentence, as follows:

$$\begin{aligned} R_{lcs} &= \frac{LCS(X, Y)}{m} \\ P_{lcs} &= \frac{LCS(X, Y)}{n} \\ F_{lcs} &= \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \end{aligned} \quad (20)$$

Where $LCS(X, Y)$ is the length of a longest common subsequence of X and Y , and $\beta = P_{lcs}/R_{lcs}$ when $\partial F_{lcs}/\partial R_{lcs} = \partial F_{lcs}/\partial P_{lcs}$. In DUC, β is set to a very big number ($\rightarrow \infty$). Therefore, the LCS-based F-measure, i.e. Equation 20, is Rouge-L.

Dist-2. Li et al. (2015) report the degree of diversity by calculating the number of distinct unigrams and bigrams in generated responses. The value is scaled by the total number of generated tokens to avoid favoring long sentences:

$$Dist(n) = \frac{Count(unique\ n-gram)}{Count(n-gram)} \quad (21)$$

C.3 Principle of Human Scoring

Details of Human Annotation. To systematically assess the model performance, we ask 8 human evaluators to rate the model responses across multiple dimensions. Evaluators are required to independently evaluate each sample in strict accordance with the pre-established criteria. We conduct cross validation of their results to avoid personal bias. The lowest and the highest scores are removed and the rest are averaged.

Evaluation Dimensions. We start with the criteria proposed by Kang et al. (2024). The human evaluation is aimed to align with the ultimate purpose of emotional support conversation, the seeker’s *satisfaction*. To achieve this, the supporter’s behavior can be further classified into the following criteria: *Acceptance*: Does the seeker accept without discomfort; *Effectiveness*: Is it helpful in shifting negative emotions or attitudes towards a positive direction; *Sensitivity*: Does it take into consideration the general state of the seeker. *satisfaction*: Does it resolve their emotional distress, leading to an overall positive perception of the conversation?

To achieve a more elaborate assessment, we consider three more dimensions addressing the generation quality:

Fluency: the level of fluency of response.

Emotion: the emotional intensity of response which could affect the seeker's emotion state.

Details of Scoring Criteria. The detailed manual scoring criteria are as follows:

- Fluency:

1: The sentence is highly incoherent, making it extremely difficult to understand and failing to convey a meaningful idea.

2: The sentence has significant incoherence issues, with only parts of it making sense and struggling to form a complete thought.

3: The sentence contains some incoherence and occasional errors, but can still convey the general meaning to a certain extent.

4: The sentence is mostly fluent with only minor errors or slight awkwardness in expression, and effectively communicates the intended meaning.

5: Perfect. The sentence is completely fluent, free of any errors in grammar, punctuation, or expression, and clearly conveys the idea.

- Emotion:

1: The emotional expression is extremely inappropriate and chaotic, not in line with the content, and may convey the wrong emotions.

2: The emotional expression has obvious flaws, either too weak or exaggerated, and is disjointed from the content.

3: The emotional expression is average. It can convey basic emotions but lacks depth and has minor issues.

4: The emotional expression is good. It can effectively convey the intended emotion with an appropriate intensity and is well integrated with the content.

5: The emotional expression is excellent. It is rich, nuanced, and perfectly matches the content, capable of evoking a strong and appropriate emotional response.

- Acceptance:

1: The response inescapably triggers emotional resistance.

2: The response is highly likely to trigger emotional resistance.

3: The response has a possibility of emotional resistance occurring.

4: The response rarely provokes emotional resistance.

5: The response has no occurrence of emotional resistance.

- Effectiveness:

1: The response actually worsens the seeker's emotional distress.

2: The response carries the risk of increasing stress levels, and this outcome varies depending on the individual user.

3: The response fails to alter the seeker's current emotional intensity and keeps it at the same level.

4: The response shows promise in calming the emotional intensity; however, it is overly complicated or ambiguous for the user to fully comprehend and utilize effectively.

5: The response appears to be highly effective in soothing the seeker's emotions and offers valuable and practical emotional support.

- Sensitivity:

1: The response renders inaccurate evaluations regarding the seeker's state.

2: The response is characterized by rash judgments, as it lacks adequate assessment and in-depth exploration of the seeker's state.

3: The response is formulated with a one-sided judgment and a limited exploration of the seeker's state.

4: The response demonstrates an understanding that only covers a part of the seeker's state.

5: The response precisely grasps the seeker's state and is appropriately tailored according to the seeker's actual situation.

- Satisfaction:

1: The response is extremely disappointing. It doesn't answer the question at all and is of no help.

2: The response is poor. It only gives a partial answer and leaves many doubts unresolved.

3: The response is average. It meets the basic requirements but isn't particularly outstanding.

4: The response is good. It answers the question clearly and provides some useful details.

5: The response is excellent. It not only answers the question perfectly but also offers valuable additional insights.

D Introduction of Baselines

We categorize the baselines into five groups, and implement most of them on the same LLM backbone unless otherwise specified.

Direct Inference.

- **Direct:** The model generates a response given the dialogue history without additional guidance.
- **Direct-Refine:** A simple two-step approach where the model first generates a response, then revises it by incorporating user feedback directly.

Prompt-based Methods.

- **Self-Refine** (Madaan, 2023): Automatically generates satisfaction-oriented feedback based on its own output, and then uses this feedback to refine its initial response.
- **Emotional Chain of Thoughts (ECoT)** (Li et al., 2024c): An emotional analogue of Chain-of-Thought prompting, which first predicts the seeker's emotion, followed by strategy and response generation guided by that emotion.
- **Skeleton of Thoughts (SoT)** (Ning et al., 2024): Generates a "skeleton" or outline of key points, and then completes each point in parallel to form the final response.
- **Tree of Thoughts (ToT)** (Yao et al., 2023): Decomposes complex problems into a tree of intermediate reasoning steps. The model explores and evaluates branches to select the best reasoning path.
- **Plan-and-Solve (PS)** (Wang et al., 2023): First generates a structured plan of subgoals or reasoning intentions, then solves each step in sequence, especially useful for multi-turn decision making.

- **FSM** (Wang et al., 2024): A prompt-based method that guides the model through a finite state machine, where transitions are triggered by current dialogue context, enabling discrete strategy control.

Finetuning-based Methods.

- **SFT:** Supervised finetuning on the training set with annotated strategy labels and response.
- **ECoT-SFT:** The finetuned version of ECoT.
- **EmoFSM** (Zhao et al., 2026): The finetuned version of FSM on ESC tasks.

Dataset-specific Methods.

- **Guyu** (Li, 2020): A BERT-based encoder-decoder model finetuned on EmpatheticDialogues with emotion and persona conditioning.
- **MoGPT** (Farahani, 2023): A transformer-based model specifically trained on DailyDialog for multi-turn response generation.
- **MPB-ESC** (Kang et al., 2024): A large-scale model built for the emotional support conversation benchmark using multi-perspective modeling and emotional strategy classification.
- **DialoGPT** (Zhang et al., 2020): A pretrained generative dialogue model evaluated on ES-Conv with task-specific tuning. We report numbers from the original paper.

Reinforcement Learning-based Methods.

- **DQN:** A Deep Q-Network that learns a policy over discrete response strategies using estimated action-values as feedback.
- **PPO:** Proximal Policy Optimization is used to optimize response selection via a reward model based on emotional alignment and coherence.
- **ArCHer** (Zhou et al., 2024b): A hierarchical RL method that learns to generate and execute high-level strategies via two-stage training with simulated user feedback.
- **DAT** (Li et al., 2024b): A method that equips pretrained language models with a planner

that treats each utterance as an action, enabling goal-directed multi-turn dialogue by training a small planner to steer generation and significantly improving performance in social simulations and strategic tasks.

For dataset-specific baselines, we directly report results from the original papers. All other methods are re-implemented using the same model backbone (unless otherwise noted) to ensure fair comparison. Hyperparameter configurations and additional training details are provided in our code repository.

E Theoretical Proofs

Connections between hierarchical MDP to SMDP. As shown in Section 2, an MDP can be defined as $M \doteq (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \gamma)$ and consider episodic tasks. In our work, we extend this traditional MDP to a hierarchical, two-level MDP, which may also be considered as a Semi-Markov decision processes (SMDP) from the perspective of a single unified time scale in previous literature (Sutton et al., 1999).

The formal MDP definition is generalized for SMDP as follows: An additional element $\mathcal{F}$ is added to the standard MDP components, resulting in SMDP as $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma, \mathcal{F})$. $\mathcal{F}$ is a function that defines the cumulative probability distribution over the number of time steps until the next controlled state. The probability that the next controlled state has been reached by time t when action a is taken in state s is written: $\mathcal{F}(t | s, a)$. It is assumed that the distribution over the number of time steps taken does not change on subsequent trials of a in s . In our work, $\mathcal{F}(t | s, a) = L$, where L denotes the max length of low-level output.

To obtain our theoretical conclusion based on previous derivations, we re-formularize our high-level MDP under the SMDP framework, where the Bellman equation becomes:

$$V^*(s) = \max_{a \in \mathcal{A}} \mathcal{R}(s, a) + \sum_{s' \in \mathcal{S}} \mathcal{T}(s, a, s') \int_0^\infty \gamma^t V^*(s') \mathcal{F}(t | s, a) dt \quad (22)$$

For simplicity of derivation, here we use s, a to represent s^H and a^H . Maximizing of $V^*(s)$ then produce the optimal strategy.

Since the state values are essentially constants when the integral is computed, $V^*(s')$ can be

pulled out of the integral, yielding:

$$V^*(s) = \max_{a \in \mathcal{A}} \mathcal{R}(s, a) + \Gamma \sum_{s' \in \mathcal{S}} \mathcal{T}(s, a, s') V^*(s')$$

where $\Gamma = \int_0^\infty \gamma^t \mathcal{F}(t | s, a) dt$ and is interpreted as a discount rate that varies with the state and action. The dynamic programming operator for a particular action is then:

$$J_a(s) = \mathcal{R}(s, a) + \Gamma \sum_{s' \in \mathcal{S}} \mathcal{T}(s, a, s') V(s')$$

and in general:

$$J(s) = \max_a \left[\mathcal{R}(s, a) + \Gamma \sum_{s' \in \mathcal{S}} \mathcal{T}(s, a, s') V(s') \right]$$

Proof of Theorem 1. Based on previous definitions, we first introduce Lemma 1 from (Sutton et al., 1999), which is originally proposed in (Parr et al., Hierarchical Control and Learning for Markov Decision Processes, Ph.D Dissertation, 1998).

Lemma 1. *The SMDP Q -learning rule converges to the optimal $Q^*(s, a)$ values if*

1. *The state and action spaces are finite.*
2. *$\sum_i \alpha_i(s, a) = \infty$ and $\sum_i \alpha_i^2(s, a) < \infty$ uniformly over s and a w.p.1.*
3. *$\text{Var}\{r\}$ is finite.*
4. *$0 < \max_{s,a} \Gamma < 1$.*

Then we prove Theorem 1 based on Lemma 1:

Proof of Theorem 1. Our work satisfies all of the above conditions, thereby allowing us to establish Theorem 1. The justifications are as follows:

1. According to the assumptions in our setting, the state space (e.g., text sequences) has a bounded length, and is therefore finite. Similarly, the action space is also finite.
2. The second condition is part of our basic assumption.
3. The reward is an integer in the range of 0 to 5, which implies that its variance is finite.
4. The discount factor γ is explicitly defined in our framework and clearly satisfies $0 < \max_{s,a} \Gamma < 1$.

Thus, the theorem is proved. $\square$

Dimensions ↓	ρ	p
Fluency	0.89	***
Acceptance	0.85	***
Effectiveness	0.81	**
Sensitivity	0.91	***
Emotion	0.88	**
Satisfaction	0.86	**

Table 12: Correlation between GPT-4o and human evaluation scores. For the significance level of p -value, we use ** to denote $p < 0.01$, and *** to denote $p < 0.001$.

F More Experimental Results

F.1 Consistency between GPT4o and human scoring

The main driven component of the reward in ToSCA is r^{sat} , which is judged by an oracle model, *i.e.*, GPT-4o. Comparing to the truly ‘golden’ evaluation, *i.e.*, human evaluation, their consistency might be critical for the experimental conclusions of this paper. To validate the reward mechanism, we conduct the correlation studies (including the Spearman coefficient ρ and the p -value of correlation slopes), with result shown in Table 12.

Table 12 shows that GPT-4o’s scores on six evaluation dimensions, including Fluency, Acceptability, Efficiency, Sensitivity, Emotion, and Satisfaction. GPT4o scores exhibit significant correlations with human judgments, with the Spearman correlation coefficients all above 0.8, and the significance level of p all below 0.01 or 0.001.

F.2 Training Curves

Figure 5 shows the training loss curve of ToSCA on DailyDialog. The $\mathcal{L}_{QL}$ decreases the fastest and remains stable, and both $\mathcal{L}_{QH}$ and $\mathcal{L}_{\pi^L}$ decrease with fluctuations. The KL divergence and intrinsic motivation value increase during the initial stages but are kept within a certain range. Importantly, r^{sat} gradually increases throughout the training process.

F.3 Human Evaluation Results

The results of human evaluations, as shown in Table 13, confirm that ToSCA’s responses align with human preferences better than baselines. To validate the inter-agreement of different human annotators, we also calculate the Cohen’s Kappa, with result of 0.47.

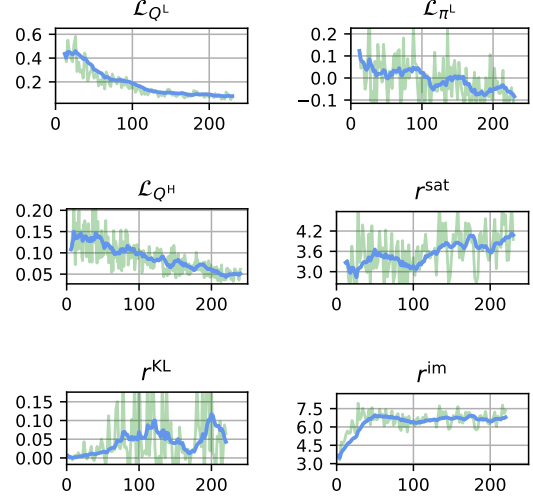


Figure 5: Training curves of ToSCA on the DailyDialog, including losses (denoted by $\mathcal{L}$) and rewards (denoted by r).

F.4 Results with More Data-Specific Baselines

Table 14 and 15 provide more completed results on DailyDialog and ESConv, with the inclusion of data-specific methods.

F.5 Detailed results of Strategies.

On the DailyDialog test set, Table 16 presents the performance of Direct and ToSCA across different dialogue strategies, which are grouped according to the ground-truth labels. It can be observed that ToSCA exhibits more balanced performance across strategies, without any notably low accuracy (e.g., 9.65). Additionally, it can be seen that the "question" strategy tends to correspond to higher B-2 scores and lower D-2 scores.

F.6 Per-strategy Results from Prediction Perspective

Table 18 presents automatic evaluation metrics grouped by predicted strategy labels rather than ground-truth labels. The overall trends are similar, with *Question* and *Inform* strategies showing relatively higher performance across metrics such as Acc, MaF1, and B-2. This indicates that precision and recall are consistent for these categories, reflecting that the model’s predictions align well with strategy patterns.

F.7 Case analysis

Typical good cases. Table 17 presents a cooking-related daily dialogue. Compared to the rigid style in the Direct group (e.g., simple repetition of the

Method	Human Annotation					
	Fluency	Acceptance	Effectiveness	Sensitivity	Emotion	Satisfaction
Direct	2.95 ± 1.41	2.60 ± 1.15	2.40 ± 0.92	2.70 ± 1.08	3.00 ± 1.34	2.60 ± 1.41
+ Direct-Refine	3.09 ± 1.25	2.73 ± 1.22	2.91 ± 1.41	2.91 ± 1.23	3.09 ± 1.16	2.84 ± 1.40
+ Self-Refine	3.10 ± 1.29	2.80 ± 1.19	2.70 ± 1.14	2.90 ± 1.03	3.15 ± 1.38	2.80 ± 1.20
+ ECoT	3.08 ± 1.02	2.83 ± 1.27	2.67 ± 1.06	3.00 ± 1.27	3.08 ± 1.29	2.83 ± 1.10
+ FSM	3.30 ± 1.32	2.90 ± 1.17	2.90 ± 1.03	3.00 ± 1.27	2.93 ± 1.19	3.00 ± 1.25
SFT	3.15 ± 1.44	2.70 ± 1.19	2.70 ± 1.20	2.90 ± 1.24	3.40 ± 1.30	2.90 ± 1.32
+ CoT	3.67 ± 1.21	3.22 ± 1.25	3.67 ± 1.26	3.56 ± 1.13	3.61 ± 1.17	3.45 ± 1.31
+ FSM	3.30 ± 1.32	2.90 ± 1.17	2.90 ± 1.03	3.00 ± 1.27	3.55 ± 1.16	3.65 ± 1.19
+ ToSCA(ours)	3.61 ± 1.26	3.86 ± 0.95	3.72 ± 1.04	3.73 ± 1.14	3.98 ± 1.21	3.91 ± 1.06

Table 13: Averaged Human evaluation of response quality on DailyDialog and ESConv.

Methods	Acc ↑	MaF1 ↑	<i>bias</i> ↓	B-2 ↑	R-L ↑	D-2 ↑
Direct	52.60	18.03	1.66	3.35	10.33	44.74
+ Direct-Refine	48.27	28.28	0.70	2.56	8.7	43.67
+ Self-Refine (Madaan, 2023)	49.76	22.15	1.18	2.4	7.75	34.01
+ ECoT(Li et al., 2024c)	38.94	29.99	0.27	1.78	6.0	55.26
+ SoT (Ning et al., 2024)	N/A	N/A	N/A	2.53	7.97	59.98
+ ToT (Yao et al., 2023)	N/A	N/A	N/A	2.52	8.84	43.19
+ PS (Wang et al., 2023)	N/A	N/A	N/A	2.60	7.76	39.73
+ FSM (Wang et al., 2024)	46.86	21.22	1.3	2.7	9.44	38.75
SFT	60.19	44.82	0.82	6.81	18.52	43.36
+ECoT	60.11	44.9	0.66	6.61	18.07	42.87
+FSM (Zhao et al., 2026)	60.03	46.02	0.55	5.85	21.77	47.43
Guyu + BigLM-12 (Li, 2020)	-	-	-	14.56	-	44.31
Guyu + BigLM-24 (Li, 2020)	-	-	-	2.93	-	61.74
MoGPT + VanillaGPT-2(354M) (Farahani, 2023)	-	-	-	-	20.8	-
DQN	54.01	50.10	0.62	4.18	13.09	59.27
DAT (Li et al., 2024b)	N/A	N/A	N/A	3.45	11.80	0.90
ArCher (Zhou et al., 2024b)	50.41	42.67	0.21	5.17	14.35	55.16
ToSCA (ours)	63.64	58.91	0.63	16.35	35.22	62.67

Table 14: Result comparisons on DailyDialog, including dataset-specific methods. The best results of each LLMs are **bolded**.

N/A: method not available for the specific metric.

-: not reported in the original work.

user’s utterances), ToSCA uses dialogue strategies more effectively to guide the user and sustain the topic. GPT-4o evaluation shows higher reward and long-term value, indicating improved conversational quality.

Bad case analysis. Despite the strong performance of ToSCA, some low-quality outputs are inevitably observed, mainly due to two reasons. First, the use of clearly inappropriate dialogue strategies leads to degraded conversational quality. Second, hallucination phenomena arising from reinforcement learning training can also impair response fidelity. As shown in Table 19, when responding to a direction inquiry, the *Commissive* strategy is evidently unsuitable, resulting in stiff and unnatural

replies. Even when the correct *Directive* strategy is selected, the model may still produce semantically unclear utterances (e.g., “Upon completion. . .”).

Methods	Acc $\uparrow$	MaF1 $\uparrow$	<i>bias</i> $\downarrow$	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
Direct	11.80	10.26	1.61	3.47	10.64	33.45
+ Direct-Refine	17.08	11.07	1.27	3.10	6.13	14.22
+ Self-Refine (Madaan, 2023)	17.58	13.61	1.92	3.34	9.71	14.61
+ ECoT(Li et al., 2024c)	15.32	10.38	1.69	3.16	10.50	30.38
+ SoT (Ning et al., 2024)	N/A	N/A	N/A	3.07	8.76	26.15
+ ToT (Yao et al., 2023)	N/A	N/A	N/A	2.65	9.81	14.2
+ PS (Wang et al., 2023)	N/A	N/A	N/A	2.81	8.27	19.12
+ FSM (Wang et al., 2024)	17.37	11.15	0.81	4.12	11.83	35.43
SFT	32.43	21.29	1.28	6.97	16.59	50.45
+ECoT	30.80	17.70	1.35	6.51	15.00	34.96
+FSM (Zhao et al., 2026)	28.00	23.70	0.41	5.88	15.30	51.48
DialoGPT-Vanilla (Liu et al., 2021)	-	-	-	5.13	15.26	-
DialoGPT-Joint (Liu et al., 2021)	-	-	-	5.00	15.09	-
DialoGPT-Oracle (Liu et al., 2021)	-	-	-	5.52	15.82	-
MPB-ESC + LLaMA2-7B (Kang et al., 2024)	-	13.73	0.77	4.98	13.09	34.74
MPB-ESC + LLaMA2-70B (Kang et al., 2024)	-	14.55	0.47	6.15	14.29	30.95
MPB-ESC + Vicuna-13B (Kang et al., 2024)	-	12.85	0.74	6.55	14.43	24.15
MPB-ESC + Mistral-7B (Kang et al., 2024)	-	12.23	0.71	4.72	12.93	25.36
MPB-ESC + Solar-10.7B (Kang et al., 2024)	-	14.17	0.87	4.79	13.53	32.36
MPB-ESC + Tulu-70B (Kang et al., 2024)	-	15.93	0.9	6.9	13.94	23.78
MPB-ESC + ChatGPT (Kang et al., 2024)	-	13.50	1.38	6.27	15.18	36.9
MPB-ESC + GPT4 (Kang et al., 2024)	-	18.38	0.90	6.47	15.18	36.92
DQN	<u>37.69</u>	<u>34.57</u>	0.59	3.59	11.74	44.14
ArCher (Zhou et al., 2024b)	24.50	19.60	0.50	5.30	13.10	54.80
ToSCA (ours)	39.26	36.85	<u>0.48</u>	<u>6.93</u>	<u>16.28</u>	<u>52.42</u>

Table 15: Result comparisons on ESConv, including dataset-specific methods. The best results are **bolded** and the second best are underlined.

N/A: method not available for the specific metric.

-: not reported in the original work.

Table 16: Per-strategy automatic metrics on DailyDialog.

	Strategy	Acc $\uparrow$	MaF1 $\uparrow$	<i>bias</i> $\downarrow$	B-2 $\uparrow$	R-L $\uparrow$	D-2 $\uparrow$
Direct	Dir.	18.47	20.35	0.81	4.40	12.73	74.96
	Inf.	41.36	52.51	0.45	14.47	29.88	70.56
	Que.	57.61	59.13	0.62	20.09	39.59	66.28
	Com.	9.65	6.01	3.95	4.25	13.07	76.18
ToSCA	Dir.	54.18	55.83	0.66	14.25	31.97	77.34
	Inf.	65.64	69.89	0.57	17.18	38.15	71.52
	Que.	58.63	62.29	0.56	19.81	40.67	65.18
	Com.	62.11	40.73	2.19	17.65	34.34	70.09

Characters	Strategy	Response	Reward	Value
User	-	Good evening, Saliva. What's that wonderful aroma from your kitchen? What are you doing now?	-	-
Assistant	Inform	I am cooking now!	3	11.7
Direct	Question	Based on your question, it seems there is a pleasant aroma coming from the kitchen area. Could you be referring to the scent of food preparation? Are you particularly interested in what is being cooked right now?	2	8.7
ToSCA	Inform	I'm cooking some fish with a coconut ginger marinade. That aroma you smell is the ginger sizzling in the pan—it's my favorite part!	5	15.1
User	-	Are you good at cooking?	-	-
Assistant	Commissive	I have studied its skills recently at a training school. I plan to run a restaurant, so I have to practise cooking!	4	10.3
Direct	Inform	Yes, I would say I have developed a solid foundation in culinary tasks over time, particularly due to frequent practice sessions and structured cooking routines.	4	7.9
ToSCA	Commissive	I've been practicing a lot lately because I want to open a small seafood restaurant someday.	4	11.9
User	-	It is a good idea.	-	-
Assistant	Question	I am steaming fish in a pressure cooker. Have you ever tried steamed fish?	4	7.4
Direct	Question	Are you saying it's a good idea because engaging in cooking activities can help improve your overall dietary quality and personal satisfaction with meals?	2	4.6
ToSCA	Question	Thank you! Have you ever tried cooking fish at home? I'd love to hear how you prepare it.	5	9.3
User	-	But the taste of steamed fish at that restaurant is not delicious. The fish is not well-done and too salty.	-	-
Assistant	Inform	That's the case. I think there is not enough time to steam it and soy sauce is too much.	4	4.0
Direct	Inform	That sounds like an unfortunate dining experience. It's possible that the restaurant staff didn't allow enough time for proper steaming, or they added too much soy sauce during preparation. You might consider exploring another restaurant next time to see if the taste is more acceptable.	3	3.0
ToSCA	Directive	It seems that restaurant's style wasn't really your thing. Next time, try adding a slice of lemon and a little salt when you steam the fish — it'll make the flavor much fresher!	5	5.0

Table 17: Case of ToSCA in DailyDialog.

	Strategy	Acc ↑	MaF1 ↑	B-2 ↑	R-L ↑	D-2 ↑
Direct	Dir.	32.14	38.22	12.43	25.66	66.85
	Inf.	49.88	54.21	15.97	32.10	62.75
	Que.	58.93	60.18	19.85	38.72	59.92
	Com.	12.47	10.05	6.12	14.90	69.34
ToSCA	Dir.	55.20	56.80	13.75	29.32	70.89
	Inf.	66.42	70.25	17.02	37.58	64.10
	Que.	61.84	63.10	19.92	40.05	66.75
	Com.	56.78	48.66	17.45	35.20	65.66

Table 18: Per-strategy automatic metrics on DailyDialog, grouped by predicted strategy. Bias is omitted as it is defined only against ground-truth labels.

User	Excuse me. Do you know where the Admiral Hotel is?
Assistant	That's over on Callaway Street, isn't it?
User	Yes, but I can't find Callaway Street.
case 1	(Commissive) I promise to make sure you walk for four blocks and then find the hotel two blocks north.
case 2	(Directive) Upon completion, initiate a northward movement for exactly two blocks to achieve destination acquisition.

Table 19: Bad cases of ToSCA responses with wrong dialogue strategies or hallucinated response.